Year 11 Mathematics

Shape and Space 6

10.5 Constructions

Constructions

STARTER · 5 MIN

Date: _____ **Title:** Constructions

LEARNING OBJECTIVES

- Construct perpendicular bisectors and perpendiculars
- Construct angle bisectors and 60°, 30° and 45° angles
- Construct triangles from three sides

STARTER *Answer in your book. No calculators.*

1. How many degrees are in a right angle?
2. Find half of 70°
3. Which tool do we use to draw arcs?
4. Find half of a 10 cm line

Starter — Answers

STARTER

Tick or correct in green pen

1. **90°**

2. $70 \div 2 = \mathbf{35°}$

3. **Compasses**

4. $10 \div 2 = \mathbf{5\ cm}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can construct perpendicular bisectors and perpendiculars	☐
SILVER	I can construct angle bisectors and 60°, 30° and 45° angles	☐
GOLD	I can construct triangles from three sides	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Constructions

INTRODUCTION · 3 MIN

Use only a ruler and compasses. Leave all your construction arcs.

Compasses

Keep the same radius for matching arcs.

Never rub out the arcs

Perpendicular bisector

Cuts a line in half at 90°.

Arcs from both ends

Angle bisector

Cuts an angle exactly in half.

Arcs from the vertex, then each arm

Accuracy

Lengths ± 1 mm, angles ± 1°.

Use a sharp pencil

Examples 1 and 2 — Perpendiculars

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint keep the compasses the same width for both arcs

Example 1

Construct the perpendicular bisector of a line AB of length 8 cm.

Example 2

Construct the perpendicular from point P to the line.

Examples 1 and 2 — Corrections

BRONZE

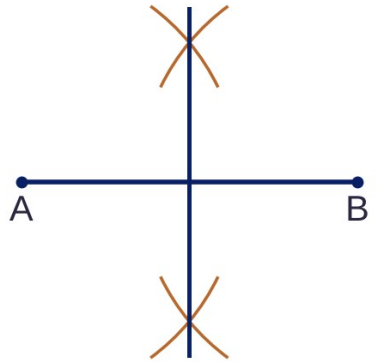
Check your copy and correct in green pen

Example 1

Compasses over half of AB

Arcs above and below from A and B

Join the crossing points



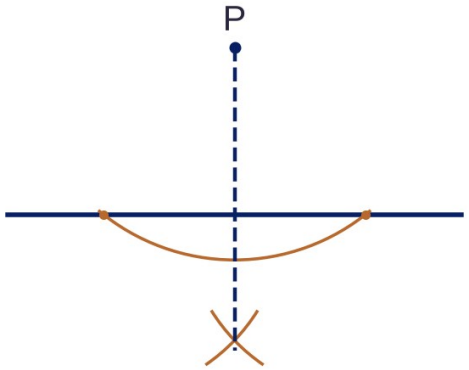
Tip: keep the same radius for all four arcs.

Example 2

Arc from P cuts the line twice

Arcs below from both points

Join P to the crossing



Tip: this is the shortest distance to the line.

Task 1

PRACTICE · 9 MIN

BRONZE

Construct each one in your book

Hint keep the compasses the same width for both arcs

1 Construct the perpendicular bisector.



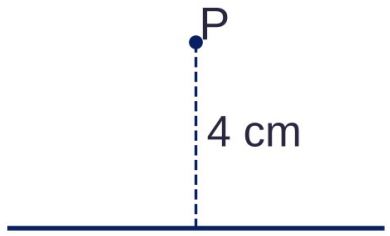
2 Construct the perpendicular bisector.



3 Find the midpoint by construction.



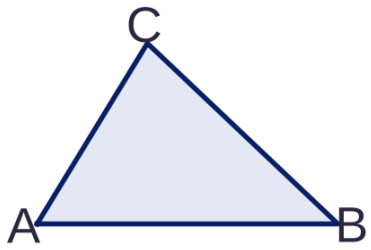
4 Construct the perpendicular from P.



5 Construct a 90° angle at the midpoint.



★ Construct the perpendicular bisector of each side.



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. Crosses AB at 3 cm, **at 90°**
2. Crosses AB at 4.5 cm, **at 90°**
3. **Midpoint is 3.7 cm from P**
4. Perpendicular line, **length 4 cm**
5. Perpendicular bisector, **angle 90°**

★ **They meet at one point**

Examples 3 and 4 — Bisecting Angles

EXAMPLES · 5 MIN

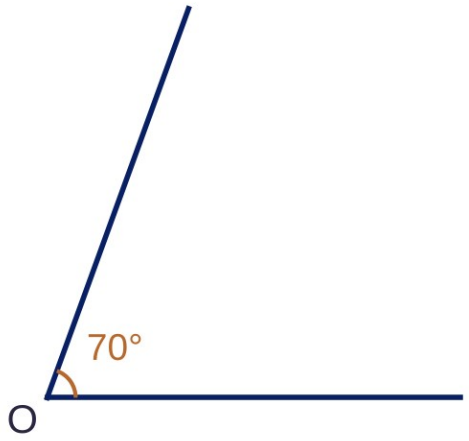
SILVER

Watch and copy into your book

Hint 60°: equilateral triangle 30°: bisect 60°

Example 3

Construct the bisector of this angle.



Example 4

Construct an angle of 60° at A.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

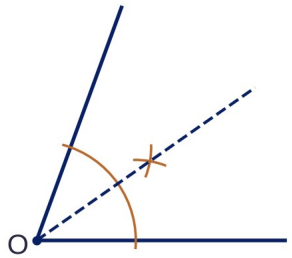
Example 3

Arc from O cuts both arms

Equal arcs from those two points

Join O to the crossing point

Each half is 35°



Tip: check with a protractor.

Example 4

Draw a line and mark A

Arc from A cuts the line at B

Same arc from B

Join A to the crossing point

Tip: this makes an equilateral triangle.

Task 2

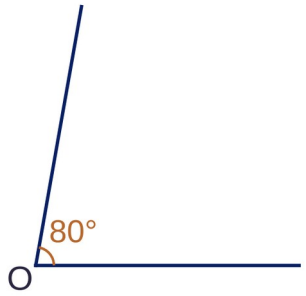
PRACTICE · 9 MIN

SILVER

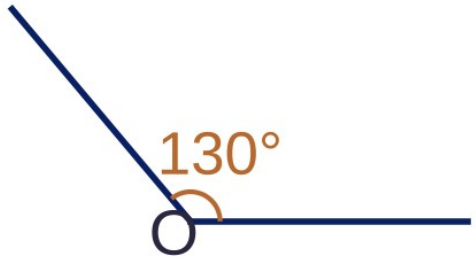
Construct each one in your book

Hint 60°: equilateral triangle 30°: bisect 60°

1 Bisect this angle.



2 Bisect this angle.



3 Construct 60° at A.



4 Construct 30° at A.



5 Construct 45° at A.



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **Each half is 40°**
2. **Each half is 65°**
3. Equilateral triangle arcs, **60°**
4. Bisect a 60° angle, **30°**
5. Bisect a 90° angle, **45°**

Examples 5 and 6 — Constructing Triangles

EXAMPLES · 5 MIN

GOLD

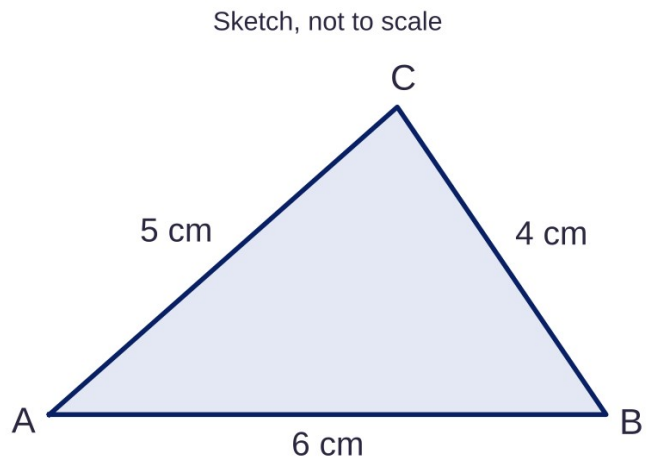
Exam-style: show all your working

Hint arcs from each end, radius = the other sides

Example 5

(3 marks)

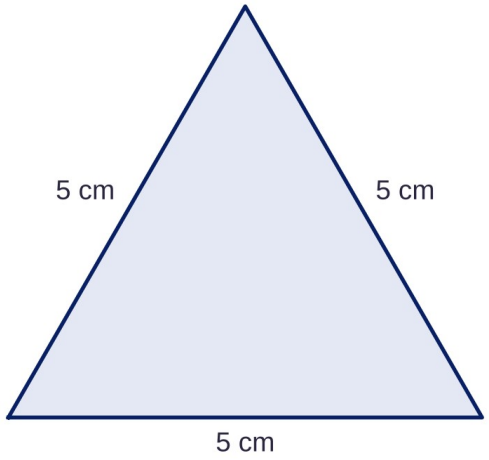
Construct triangle ABC accurately.



Example 6

(2 marks)

Construct this equilateral triangle.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

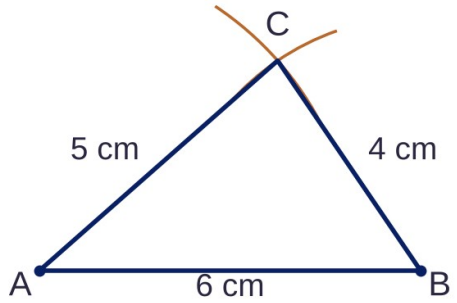
Example 5

Draw AB = 6 cm

Arc of radius 5 cm from A

Arc of radius 4 cm from B

Join A and B to the crossing point



Tip: leave your arcs showing.

Example 6

Draw a 5 cm line

5 cm arcs from both ends

Each angle is 60°

Tip: all sides are 5 cm.

Task 3

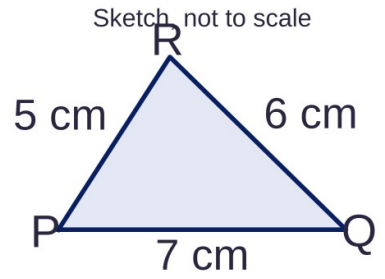
PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

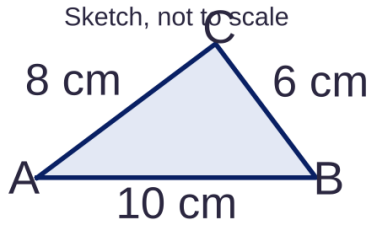
Hint arcs from each end, radius = the other sides

1 Construct triangle PQR.



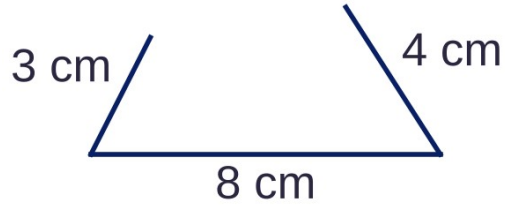
(3 marks)

2 Construct it. What type of triangle?



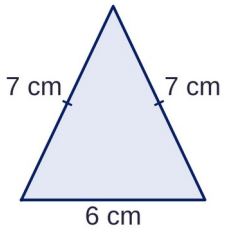
(3 marks)

3 Why can this not be constructed?



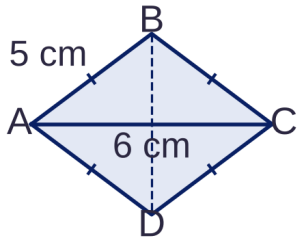
(2 marks)

4 Construct it, then bisect the top angle.



(3 marks)

5 Construct the rhombus. Measure BD.



(3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. Arcs 5 cm from P and 6 cm from Q, **check ± 1 mm**
2. $6^2 + 8^2 = 10^2$, **right – angled**
3. $3 + 4 < 8$, **the arcs do not meet**
4. Bisector meets the base at 3 cm, **at 90°**
5. Two 5, 5, 6 triangles, BD = **8 cm**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE What does a perpendicular bisector do?

SILVER An angle of 110° is bisected. What size is each half?

GOLD Can you construct a triangle with sides 2 cm, 3 cm and 6 cm?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **Cuts a line in half at 90°**

SILVER $110 \div 2 = \mathbf{55°}$

GOLD $2 + 3 < 6$, **No**

Exam Practice — Perpendiculars

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint keep the compasses the same width for both arcs

Question 1

(2 marks)

Use ruler and compasses to construct the perpendicular bisector of AB.

Show all your construction lines.

Diagram NOT accurately drawn

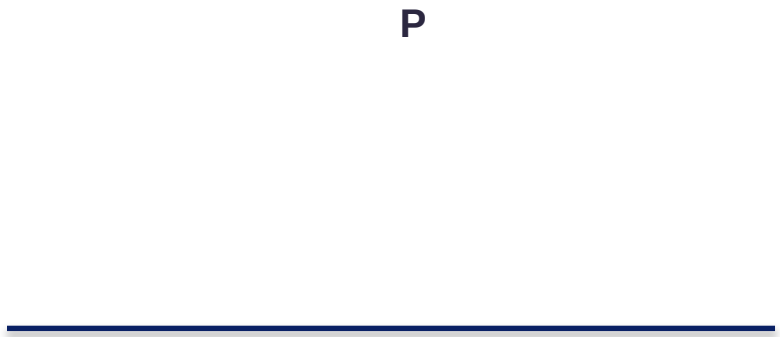


Question 2

(2 marks)

Use ruler and compasses to construct the perpendicular from P to the line.

Diagram NOT accurately drawn



Exam Practice — Bisecting Angles

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

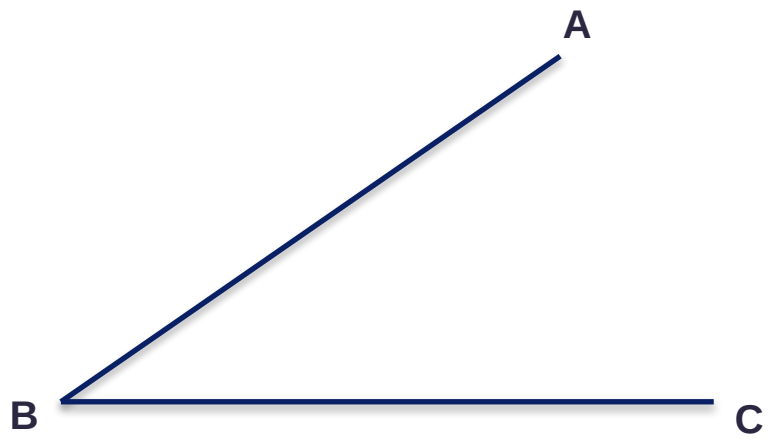
Hint 60°: equilateral triangle 30°: bisect 60°

Question 3

(2 marks)

Use ruler and compasses to bisect angle ABC.

Diagram NOT accurately drawn



Question 4

(3 marks)

Construct an angle of 30° at A.

Show all your construction lines.

Diagram NOT accurately drawn



Exam Practice — Constructing Triangles

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint arcs from each end, radius = the other sides

Question 5

(3 marks)

Use ruler and compasses to construct triangle ABC with AB = 8 cm, BC = 6 cm and AC = 7 cm.

AB has been drawn for you.

Diagram NOT accurately drawn



Question 6

(2 marks)

Explain why a triangle with sides 3 cm, 4 cm and 8 cm cannot be constructed.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

equal arcs from A and from B — M1

that cross above and below AB

line through both crossings — A1

Tip: open the compasses more than half of AB.

Question 2

arc from P cuts the line twice — M1

equal arcs from those points

line from P to the crossing point — A1

Tip: leave all your arcs showing.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

arc from B cuts both arms	M1
equal arcs from those points	
line from B through the crossing	A1

Tip: check: both halves are the same size.

Question 4

construct 60° (equilateral arcs)	M1
bisect the 60° angle	M1
30° angle with arcs shown	A1

Tip: 30° is half of 60°.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

arc radius 7 cm from A	M1
arc radius 6 cm from B	M1
triangle drawn with arcs shown	A1

Tip: ±2 mm on each length.

Question 6

3 + 4 = 7	M1
7 < 8, so the arcs do not meet	A1

Tip: short sides must beat the long side.